

Muhammad Shahzeb Ali

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Education

Doctor of Philosophy, Mathematics

Ohio University, Athens, OH, USA

2021 – Expected May 2026

Master of Science, Statistics

International Islamic University, Islamabad, PAK

2019 – 2021

Bachelor of Science, Statistics

Quaid-I-Azam University, Islamabad, PAK

2014 – 2018

Projects

Hybrid Statistical-Deep Learning Time Series Model

2025 – Present

- **Key Skills:** TensorFlow, Transformer Architecture, GARCH Modeling, MLflow, Keras Tuner
- Designing an innovative pipeline that integrates a statistical GARCH model for volatility feature extraction with a deep learning Transformer encoder and MLP for temporal forecasting.
- Engineering a novel feature set by synthesizing raw input data with conditional volatility to better capture complex, non-linear market patterns.
- Implementing and validating model components in TensorFlow to construct an end-to-end trainable system, utilizing MLflow for experiment tracking and Keras Tuner for hyperparameter optimization.

Credit Risk Classifier

2024 – 2025

- **Key Skills:** Pytorch, Optuna, Imbalance Classification
- Developed a PyTorch neural network to predict loan repayment status, achieving 96.69% accuracy and 94.87% macro F1-score with a robust 0.8928 F1-score for high-risk loans. Engineered a preprocessing pipeline for 59 features, addressing class imbalance and missing values. Optimized model performance using Optuna for hyperparameter tuning.

Professional Experience

Ohio University, Athens, OH, USA

2021 – Present

Graduate Research & Teaching Assistant

- **Machine Learning Research:** Conducted research on molecule featurization and transformer-based architectures for molecular property prediction, developing strong foundational skills in deep learning model design and implementation.
- **Instruction & Leadership:** Taught and designed curriculum for 8+ undergraduate courses in Calculus, Statistics, Probability, and Differential Equations to classes of 30+ students.
- **Data-Driven Instruction:** Analyzed student performance data to identify knowledge gaps and adapt teaching strategies, improving overall class comprehension and exam scores.

Johns Hopkins University, Center for Talented Youth

Summers 2022 – 2025

Instructor

- Instructed accelerated courses in **Cryptology**, **Mathematics of Money**, and **Probability** for high achieving students.
 - Developed and delivered engaging course materials, including lectures, problem sets, and exams, focused on practical applications of quantitative theory.
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